INCH-POUND

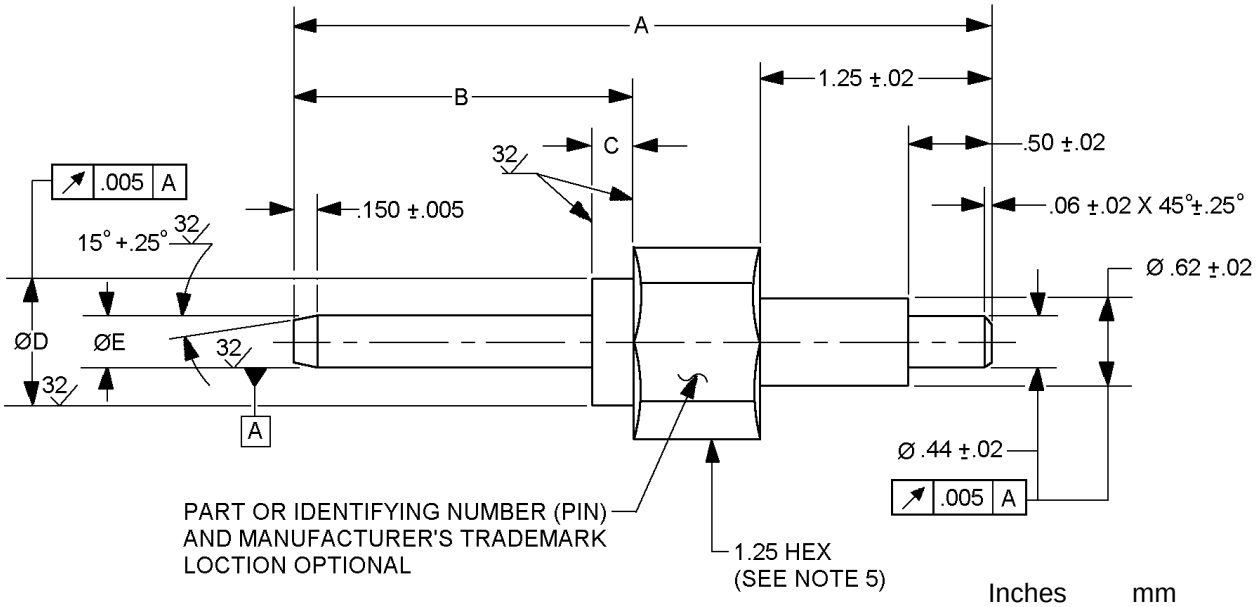
MS27638E
22 October 2009
SUPERSEDING
MS27638D
29 September 2000

DETAIL SPECIFICATION SHEET

MANDREL SLEEVE SETTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83296.




| Inches | mm |
|---|---|
| .005 | 0.13 |
| .02 | 0.51 |
| .06 | 1.52 |
| .44 | 11.18 |
| .50 | 12.70 |
| .62 | 15.75 |
| 1.25 | 31.75 |
| 1.50 | 38.10 |

FIGURE 1. Mandrel sleeve setting dimensions and configuration.



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| PIN | A ±.02 inches (cm) | B ±.02 inches (mm) | C ±.005 inches (mm) | D ±.005 inches (mm) | E | |
|---|---|---|---|---|---|---|
| | | | | | min inches (mm) | max inches (mm) |
| MS27638-4C | 3.37 (8.56) | 1.12 (28.45) | .325 (8.26) | .475 (12.07) | .225 (5.72) | .230 (5.84) |
| MS27638-6C | 3.75 (9.53) | 1.50 (38.10) | .300 (7.62) | .605 (15.37) | .322 (8.18) | .334 (8.48) |
| MS27638-8C | 4.25 (10.80) | 2.00 (50.80) | .355 (9.02) | .705 (17.91) | .440 (11.18) | .445 (11.30) |
| MS27638-10C | 4.25 (10.80) | 2.00 (50.80) | .295 (7.49) | .870 (22.10) | .575 (14.61) | .580 (14.73) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances shall be ±.005 inch (0.13 mm).
4. Surface roughness, in accordance with ASME B46.1, shall not exceed 125 μin $R_a$, unless otherwise specified.
5. For hex, use commercial stock size tolerance in accordance with SAE-AMS2241.
6. Dimensioning and tolerancing are in accordance with ASME Y14.5M.
7. Unless otherwise specified, radius or break all corners to .005 +.005/-.000 inch (0.13 +0.13/-0.00 mm).
8. All burrs and slivers shall be removed.

FIGURE 1. Mandrel sleeve setting dimensions and configuration - Continued.

REQUIREMENTS:

Dimensions and configurations: The design, construction, and physical dimensions shall be in accordance with MIL-DTL-83296 and figure 1, in case of conflict between this specification sheet and MIL-DTL-83296, this specification sheet shall govern.

This part for use with hose assemblies in accordance with MIL-DTL-32330.

Material: Material shall be tool steel alloy UNS T41902, type 2 in accordance with ASTM A681.

Hardness: Hardness shall be Rockwell C 58 to 60.

Finish: Finish shall be black oxide in accordance with SAE-AMS2485.



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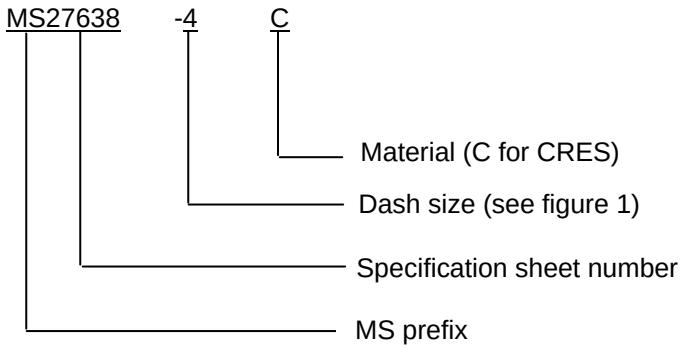
PIN example:




PIN example: MS27638-4C indicates a sleeve setting mandrel 1/4 inch, tool steel alloy.

Changes from previous issue. Marginal notations are not used in this revision to identify changes due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-83296, this document references the following:

| | |
|---|---|
| MIL-DTL-32330 | ASTM A681 |
| ASME B46.1 | SAE-AMS2241 |
| ASME Y14.5M | SAE-AMS2485 |

## CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Review activities:
Navy - MC, SA
Air Force - 71

Preparing activity:
DLA - CC

(Project 4730-2008-069)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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